

STAT41120 - Machine Learning & AI

Assignment - Report

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1. Introduction

This report addresses the classification of 19 daily and sports activities from motion sensor data collected across 8 subjects wearing 5 body-mounted sensor units. Each observation is a 5-second signal segment represented as a 125×45 matrix (125 time steps at 25 Hz, 45 sensor channels). Two deep learning architectures are implemented, compared, and evaluated: a fully-connected Deep Neural Network (DNN) and a 1D Convolutional Neural Network (CNN). The training set contains 8170 signal segments (430 per class, perfectly balanced), and the test set contains 950 segments (50 per class).

2. Data Preprocessing

For the DNN, each 125×45 signal matrix was flattened into a 5625-dimensional feature vector. Features were then standardised column-wise using training-set means and standard deviations, with the same statistics applied to the test set to prevent data leakage. For the 1D CNN, the original 3D structure ($n \times 125 \times 45$) was preserved to retain temporal ordering. Standardisation was performed per-channel across all training samples and time steps. Labels were converted to one-hot encoded vectors over 19 classes. A 15% stratified validation split (1226 observations) was drawn from the training data, with the same split used for both models to ensure a fair comparison.

3. Model 1 : DNN

3.1 Architecture and Motivation

The DNN serves as a strong baseline that treats all 5625 features as independent inputs with no assumed spatial or temporal structure. The architecture consists of three hidden layers with 512, 256, and 128 units respectively, forming a funnel that progressively compresses the high-dimensional input into increasingly abstract representations. Each of the first two hidden layers is followed by batch normalisation (to stabilise training and reduce internal covariate shift) and dropout at rate 0.4 (for regularisation). The third hidden layer uses dropout at rate 0.3. The output layer has 19 units with softmax activation, producing a probability distribution over activity classes.

3.2 Architecture Summary

Layer	Output Shape	Parameters
Dense (512, ReLU)	(None, 512)	2,880,512
BatchNorm + Dropout(0.4)	(None, 512)	2,048
Dense (256, ReLU)	(None, 256)	131,328
BatchNorm + Dropout(0.4)	(None, 256)	1,024
Dense (128, ReLU) + Dropout(0.3)	(None, 128)	32,896
Dense (19, Softmax)	(None, 19)	2,451

3.3 Training Configuration

A grid search over learning rate {0.001, 0.0005}, batch size {32, 64}, and dropout {0.3, 0.4} was conducted. The best configuration was lr = 0.001, batch size = 32, and dropout = 0.4. The model was compiled with the Adam optimiser and categorical crossentropy loss. Training ran for up to 50 epochs using early stopping (patience = 7, monitoring validation loss, restoring best weights). The final validation accuracy was 96.82% with a validation loss of 0.069.

4. Model 2 : CNN

4.1 Architecture and Motivation

Unlike the DNN, the 1D CNN preserves the temporal structure of the input, taking each observation as a sequence of 125 time steps with 45 channels. This allows convolutional filters to learn local temporal patterns (e.g., periodic motion in walking, sudden changes in jumping) directly from the raw signal. The architecture uses three convolutional blocks with increasing filter counts (64, 128, 256) and kernel sizes of 5, 5, and 3. Each block includes batch normalisation, and the first two apply max pooling (pool size 2) to halve the temporal dimension. Global average pooling reduces the output to a 256-dimensional vector, followed by a 128-unit dense layer with dropout (0.3 and 0.2) before the 19-class softmax output.

4.2 Architecture Summary

Layer	Output Shape	Parameters
Conv1D (64, kernel=5, ReLU)	(None, 121, 64)	14,464
BatchNorm + MaxPool(2)	(None, 60, 64)	256
Conv1D (128, kernel=5, ReLU)	(None, 56, 128)	41,088
BatchNorm + MaxPool(2)	(None, 28, 128)	512
Conv1D (256, kernel=3, ReLU)	(None, 26, 256)	98,560
BatchNorm + GlobalAvgPool	(None, 256)	1,024
Dropout(0.3) + Dense(128, ReLU)	(None, 128)	32,896
Dropout(0.2) + Dense(19, Softmax)	(None, 19)	2,451

4.3 Training Configuration

A grid search over learning rate {0.001, 0.0005}, batch size {32, 64}, and dropout {0.2, 0.3} was conducted. The best configuration was lr = 0.0005, batch size = 64, and dropout = 0.3. The model was compiled with the Adam optimiser and categorical crossentropy loss. Training ran for up to 50 epochs using early stopping (patience = 7, monitoring validation loss, restoring best weights). The final validation accuracy was 99.51% with a validation loss of 0.014.

5. Model Comparison

Metric	DNN	CNN
Validation Accuracy	96.82%	99.51%
Validation Loss	0.069	0.014
Total Parameters	~3.05M	~191K

The 1D CNN outperforms the DNN on both validation accuracy (+2.69 percentage points) and validation loss (nearly 5× lower). Notably, the CNN achieves this with substantially fewer parameters (~191K vs ~3.05M), demonstrating superior parameter efficiency. The DNN must learn temporal relationships purely from the flattened 5625-dimensional input, which discards all ordering information. In contrast, the CNN's convolutional filters explicitly

